

A spatially and temporally refined dataset of CONUS pluvial National Flood Insurance Program claims

Yuya Kawakami¹ and Tom Corringham²

¹Department of Computer Science, University of California, Davis, Davis, CA

²Scripps Institution of Oceanography, University of California, San Diego, La Jolla, CA

Abstract

The National Flood Insurance Program (NFIP) Redacted Claims dataset is the primary public record of flood losses in the United States, but two limitations of the raw data constrain its use for location- and date-sensitive analysis. First, locations are deliberately obscured to protect policyholder privacy. Each claim carries only a census block group, a ZIP code, and a latitude/longitude rounded to one decimal degree, rather than an exact address. Second, for some claims tagged as rain-driven (pluvial) flooding, the reported loss date does not coincide with any local rainfall event, likely reflecting reporting delays or processing errors. Here we present a cleaning pipeline and derived dataset that (1) converts all dollar amounts to a constant reference year, (2) combines the three location signals into a single bounded polygon per claim using a boundary-matching approach validated against the data itself, and (3) replaces likely erroneous dates for pluvial claims with the most plausible nearby day of qualifying rainfall, identified by checking each claim’s local rainfall record against NOAA’s Analysis of Record for Calibration (AORC) precipitation data. No independent ground truth exists for a claim’s true flood-triggering date, so this adjustment is best understood as a plausibility-improving heuristic rather than a verified correction. The pipeline is fully scripted and reproducible from public sources, with all refinement choices exposed as configurable parameters.

Background & Summary

The NFIP, administered by the Federal Emergency Management Agency (FEMA), is the principal source of insured flood-loss data in the United States and is widely used in flood-risk, climate, and disaster-economics research [6, 8–11, 13, 14]. FEMA publishes a redacted extract of individual claims (OpenFEMA’s *FIMA NFIP Redacted Claims v2*) [5], which is publicly available, freely accessible, and regularly updated. However, redaction leaves several properties of the raw records unsuitable for location or date-sensitive analysis without additional processing.

Location information is redacted for privacy. No street address is provided. Instead, each record is accompanied by the claim’s census block group FIPS code, its reported ZIP code, and a latitude/longitude pair rounded to one decimal degree. These constitute three coarse approximations to the claim’s true location, none of which alone constrains it tightly. Past work using redacted NFIP claims has often relied on this rounded latitude and longitude alone as a proxy for claim location. However, analyses that would benefit from more precise location information require higher spatial resolution than this proxy affords, particularly for the understudied case of pluvial flooding.

Reported loss dates are also imperfect for a subset of claims [11]. NFIP identifies pluvial (rain-driven) flooding claims via `causeOfDamage == "4"`. However, for these claims, the reported `dateOfLoss` does not always coincide with a day of actual rainfall at the claim’s location. For instance, in our subset of NFIP claims, to be described below, around 7.5% of pluvial claims have a maximum hourly accumulated precipitation of 0.0mm over the reported date of loss with reference to the NOAA Analysis of Record for Calibration

(AORC) archive [4]. This pattern is more consistent with reporting delays, adjuster lag, or transcription error than with the true date of the flood-inducing event.

To advance the modeling of pluvial flood incidence, risk, and associated damages, this paper describes an open, reproducible pipeline that addresses both problems using only public data, namely FEMA’s own claims extract, the Federal Reserve Economic Data (FRED) personal consumption expenditures price index, U.S. Census Bureau cartographic boundary files, and NOAA’s AORC v1.1 archive [4].

The resultant dataset is thus a spatially and temporally refined dataset of pluvial NFIP claims, where we associate each reported pluvial claim with a polygon that bounds the claim’s true location and, for claims with a loss date on or after AORC coverage begins (1979-02-01) and evaluable precipitation data, a corrected date of loss. We further design the dataset construction script so that it exposes pertinent refinement parameters that can be tuned to different assumptions and needs of the downstream tasks. Both the dataset and the pipeline code that produced it are released under an open license.

Methods

The pipeline processes FEMA’s public claims extract in three ordered stages, namely inflation adjustment, spatial triangulation, and pluvial date correction. Each stage is implemented as an independent, re-runnable script. The full sequence is defined in `src/run_pipeline.sh` in the code repository (see Code Availability).

Input data

- **FEMA OpenFEMA** — FIMA NFIP Redacted Claims v2 [5], snapshot retrieved 2026-08-13. This is provided in the public domain by the Federal Emergency Management Agency (FEMA) with no authentication required. Because OpenFEMA updates this extract on an ongoing basis, later retrievals will not reproduce this exact claim count.
- **FRED** (Federal Reserve Bank of St. Louis) — Personal consumption expenditures price index, series DPCERD3Q086SBEA [12]. This is provided in the public domain as a CSV file with no authentication required.
- **U.S. Census Bureau** — Cartographic boundary shapefiles, block groups and ZIP Code Tabulation Areas (ZCTAs). This is made available per decadal census. Its matching strategy with the NFIP claims is configurable as flags, with the default behavior below. This is provided in the public domain with no authentication required.
- **NOAA** — Analysis of Record for Calibration (AORC) v1.1 [4], 30 arc-second (~ 800 m) grid, hourly. This is made available via Amazon Web Services (AWS) at `s3://noaa-aws-aorc-v1-1-1km`. No authentication is required.

Scope. This dataset covers the 48 contiguous U.S. states and the District of Columbia (CONUS). NFIP claims also exist for Alaska, Hawaii, Puerto Rico, and other U.S. territories. These are excluded by design, not as an incidental side effect of processing, because both the Census cartographic boundary conventions and AORC’s grid extent treat CONUS as a distinct product from these territories. The released file is further restricted to claims coded `causeOfDamage == "4"`, matching the pluvial focus.

Table 1 records the claim count remaining after each of these restrictions, from FEMA’s inflation-adjusted extract through to the triangulation output. Entering triangulation also requires both `censusBlockGroupFips` and `reportedZipCode` to be non-null, regardless of whether a latitude/longitude pair is present, a stricter condition than requiring only one of the three location signals to be available.

Step	n claims remaining	Δ from previous step	% dropped
Start, FEMA extract, all causes of damage	2,721,780	—	—
Keep pluvial claims only (<code>causeOfDamage == "4"</code>)	1,206,336	−1,515,444	55.68%
Drop claims missing <code>censusBlockGroupFips</code> or <code>reportedZipCode</code>	1,163,030	−43,306	3.59%
Drop claims outside CONUS (AK, HI, PR, other territories)	1,155,473	−7,557	0.65%
Drop claims whose triangulated sources fail to overlap (empty intersection)	1,149,288	−6,185	0.54%

Table 1: Claim count remaining after each filtering and processing step, from FEMA’s inflation-adjusted extract through triangulation, for the pluvial CONUS population this dataset covers, based on the FEMA OpenFEMA NFIP claims snapshot retrieved 2026-08-13.

Stage 1 — Inflation adjustment

Every dollar-denominated claim field (paid amounts, coverage limits, building/contents value, replacement cost) is rescaled to a fixed target year’s dollars using the FRED PCE price index, matched to each claim’s loss quarter. The default target year is 2021 and is configurable.

Stage 2 — Spatial triangulation (*Spatial refinement*)

Each claim carries three independent, imprecise location signals, namely a census block group FIPS code, a ZIP code, and a rounded latitude/longitude. Triangulation intersects whichever of these signals are available into a single polygon per claim, yielding a strict upper bound on the claim’s true location that is typically substantially smaller than the region implied by any single source alone.

Boundary assignment. Here and throughout the rest of this paper, a *release* means a specific dated edition of a Census boundary shapefile, for example the 2010 or 2020 block-group shapefile, and is unrelated to the release of this dataset itself. Census block-group and ZCTA boundaries are redrawn each decennial cycle, so multiple releases exist for a given geography. Rather than matching each claim to the boundary release nominally in effect at its loss year, boundary-release assignment is a run-time choice, selected independently for block group and ZCTA via a strategy parameter with four options.

- **default** — for block group, the 2010 release before a configurable cutover year and the 2020 release from that year on. For ZCTA, equivalent to **closest** below, selecting whichever configured ZCTA release’s year is numerically nearest the loss year, including loss years before 2000.
- **closest** — whichever configured release’s year is numerically nearest the claim’s loss year, ties broken toward the newer release.
- **most_recent** — always the newest configured release, regardless of loss year.
- **drop** — do not use that source.

The released dataset uses **default** for both block group and ZCTA. Official block group shapefiles do not exist nationwide before 1990, and ZCTA shapefiles exist only from the 2000 census onward. Under **default**, block group is still attempted against the 2010 release regardless of how far the loss year precedes 1990, and ZIP is matched to whichever configured ZCTA release is numerically closest to the loss year, including loss years before 2000, where the 2000 release is the closest available (See *Technical Validation*).

Once a *release* is selected, or unselected in the case of **drop**, we compute the spatial intersection of the boundaries (i.e. $\text{claim polygon} = \bigcap_{i=1}^n \text{polygon}_i$, where n is the number of contributing sources). In the worst case, we fall back to the 0.1×0.1 degree box defined by the rounded latitude/longitude, but in almost all cases the intersection of the three sources is a substantially smaller polygon. Our achieved spatial refinement

is, however, not uniform in space and significantly favors urban regions with dense block-group and ZIP coverage over rural regions with sparse coverage.

Stage 3 — Pluvial date correction (*Temporal refinement*)

For each claim coded `causeOfDamage == "4"`, the reported `dateOfLoss` is checked against whether it coincides with a day of meaningful precipitation (maximum hourly precipitation exceeding a threshold, default 7.62 mm) at the claim's triangulated location. Peak hourly intensity, rather than daily-accumulated rainfall, is used as the criterion because pluvial flooding is characteristically caused by localized, intense rainfall and is conventionally assessed using local hourly rainfall records [7]. Pluvial flooding occurs when rainfall exceeds local drainage capacity within the short time it takes runoff to reach a drain or low point, so its risk is governed by short-duration rainfall intensity rather than by rainfall accumulated over a full day.

To define this threshold, we refer to the NOAA AORC hourly accumulated precipitation product. If the reported date's precipitation does not exceed the defined threshold, a window of nearby days (default ± 7) is searched for the day with the daily maximum hourly accumulated precipitation, and if that day clears the threshold it is treated as the corrected date. Within the window, the corrected date is chosen as the day with the greatest precipitation rather than the day nearest the reported date. This follows from a monotonic prior between precipitation intensity and pluvial flood risk. Among the days in the window that clear the qualifying threshold, the day with the highest precipitation is taken as the most plausible candidate under this prior, independent of which candidate day is closer in time to the reported date. No independent ground truth on the claim's true flood-triggering date exists to confirm this choice, so this ranking should be read as an assumption the pipeline makes explicit, not as a validated identification of the true event day. The original `dateOfLoss` is never overwritten. The corrected value and an explicit status code are added as new columns, letting downstream users decide whether to apply the correction.

The precipitation threshold used to judge whether a day counts as meaningful rainfall is configurable, either as a single constant or as a spatial field. This should reflect the prior beliefs, assumptions, and needs of the downstream tasks. A spatial field may be useful, as there is no principled reason a claim in an arid region and a claim in a persistently wet region should require the same amount of rainfall to qualify as *meaningful*.

- **Constant** — the default, a single value applied everywhere. The script defaults to 7.62 mm (0.30 in), the threshold the American Meteorological Society's Glossary of Meteorology uses to define *heavy* rainfall intensity [1], but this is configurable at run time.
- **Spatial field** — a user-supplied NetCDF raster sampled at each claim's exact location, giving a spatially varying threshold without requiring the raster to share AORC's grid or resolution.

Day boundaries are evaluated in each claim's own local time zone, resolved geographically per claim location, rather than a single national reference clock, and daylight-saving transitions are handled. Precipitation at each claim's triangulated polygon centroid is computed by bilinear interpolation of the four surrounding AORC grid pixels. This centroid sample is a single point estimate and does not average over the polygon's extent, so it may be less representative for the largest or most irregularly shaped triangulated polygons (Figure 1) than for the small, near-point polygons most claims resolve to. Because the median triangulated polygon (1.17 km², Figure 1) is comparable in scale to a single AORC grid cell (~ 0.8 km $\times$ 0.8 km, ~ 0.64 km²), centroid sampling introduces little additional approximation beyond AORC's native spatial resolution for most claims, and the limitation is concentrated in the right tail of large or irregularly shaped polygons.

AORC coverage for CONUS begins 1979-02-01. Pluvial claims before this date, and claims for which AORC has no data at the triangulated location (e.g. outside the grid extent, or a small land mass whose surrounding grid points are all ocean pixels), are not evaluated for date correction. These claims are retained in the released file rather than dropped. `correctedDateOfLoss` falls back to the originally reported `dateOfLoss`, and `pluvialCorrectionStatus` is set to `before_aorc_coverage` or `no_aorc_data_in_window` accordingly (see Data Records).

Data Records

The dataset is released as a single columnar file (`claims_pluvial_corrected.parquet`, Apache Parquet format, Snappy-compressed) at claim-level granularity, comprising 1,149,288 rows and 103 columns, one row per released claim, totaling approximately 1.7 GB on disk. All original columns from the FEMA OpenFEMA `FimaNfipClaims` extract are preserved unchanged (see FEMA’s data dictionary for definitions). The columns added by this pipeline are as follows.

Added by inflation adjustment.

- `{field}Real2021` for each of 15 dollar-denominated fields (paid amounts, coverage limits, building/contents value, replacement cost), rescaled to 2021 dollars.
- `amountPaid / amountPaidReal2021` — sum of the three `amountPaidOn*Claim` fields.
- `damageAmount / damageAmountReal2021` — `buildingDamageAmount + contentsDamageAmount`.

Added by triangulation.

- `geometry` (polygon, EPSG:5070 Albers Equal Area) — the intersection of whichever of {block-group boundary, ZCTA boundary, latitude/longitude box} were available and spatially validated for the claim. Claims with zero contributing sources, or whose sources fail to overlap one another, are excluded from the released file.
- `n_geometry_sources` (1–3) and `geometry_sources` (e.g. "block_group+zip+latlon") — record which sources contributed.
- `block_group_vintage_used` and `zip_vintage_used` — the boundary-release year (e.g. 1990, 2000, 2010, or 2020) selected for the claim by the block-group and ZIP release-matching strategies, respectively (see Methods). The two need not agree, even when both strategies are left at `default`.
- `block_group_match_status` — one of `not_found`, `validated`, `unvalidated_no_latlon`, `spatially_inconsistent`, `dropped_by_strategy`, or `no_vintage_selected`.
- `zip_match_status` — the same status values as `block_group_match_status`

Added by pluvial date correction.

- `correctedDateOfLoss` — the reported date if it already coincided with meaningful precipitation, the corrected date if a correction was made, or the original date otherwise (always populated).
- `pluvialCorrectionStatus` — one of `before_aorc_coverage`, `accepted_as_reported`, `corrected`, `no_qualifying_precip_in_window`, or `no_aorc_data_in_window`.
- `reportedDateMaxPrecipMm` — the maximum hourly precipitation (mm) found at the claim’s location on the originally reported `dateOfLoss`, before any correction. This is the value checked against the threshold to decide whether correction is attempted at all.
- `pluvialCorrectionMaxPrecipMm` — the maximum hourly precipitation (mm) found at the claim’s location on the date `correctedDateOfLoss` refers to.

`geometry` uses EPSG:5070 (meters). The original FEMA `latitude / longitude` fields remain in WGS84 (EPSG:4326) degrees, unchanged.

Technical Validation

Boundary-release assignment

This subsection checks whether the boundary release a claim is matched against actually corresponds to a plausible location for that claim, using each claim’s own reported coordinates as an independent check on the string-based GEOID match, rather than assuming a code match alone means the claim is correctly located. Throughout this subsection, each claim’s reported latitude and longitude, rounded by FEMA to

one decimal degree, is treated as ground truth for the claim’s true location, giving a 0.1×0.1 degree box the claim must fall within. This treatment assumes the reported latitude/longitude is itself an accurate proxy for the claim’s true location. That assumption is not independently verified here, and if the reported coordinates carry their own location error, the checks below inherit it rather than detect it.

For a given release, *Match* is the raw rate at which a claim’s block-group or ZIP code string matches a GEOID in that release’s shapefile. *Valid* is computed only over the matched claims, and reports the share of those matches whose polygon also overlaps the claim’s ground-truth latitude/longitude box, exactly the spatial-consistency check described in Methods. A high Match rate with a low Valid rate would mean the GEOID exists in that release but points to the wrong place. A high Valid rate confirms that when a release matches at all, it matches to a polygon consistent with the claim’s own reported coordinates. Because the claim’s rounded latitude/longitude is itself one of the three sources intersected into the final triangulated polygon (see Methods), Valid measures agreement between the block-group or ZIP source and the lat/lon source, not agreement against an external ground truth independent of the triangulation inputs. A high Valid rate therefore establishes internal consistency among the pipeline’s own location signals, not the accuracy of any one of them against a source outside the pipeline.

Block-group boundaries. Restricted to the released (pluvial, `causeOfDamage == "4"`) claim population, matching `censusBlockGroupFips` against the official 2010 decennial block-group release yields a 93.3–100% match rate for every loss year from 1978 through 2015, remaining the better-matching release through 2019 even as its raw match rate declines to 84.3–87.5% over 2016–2019. This indicates FEMA’s geocoding assigns a recent-release boundary to each record at processing time rather than using the historical loss-year boundary. In absolute terms the transition to the newer (2020) release is not sharp. The raw match rate against 2020 climbs gradually with loss year rather than jumping at a single point, reaching only 83.2% in 2020 itself before climbing to 100% by 2026. Ranked against each other, however, the 2010 release is the strictly better-matching release for every loss year from 1978 through 2019, and the 2020 release becomes the strictly better-matching release for every loss year from 2020 through 2026, with no exceptions across this population (Table 4).

This clean crossover is evidence supporting a strong working hypothesis, namely that FEMA’s geocoding service itself assigns the 2010 release to claims with loss year before 2020 and the 2020 release to claims with loss year 2020 or later, even though general match-quality variation keeps the raw match-rate curve gradual rather than a step. This working hypothesis is independently supported by FEMA’s own data dictionary, which describes how `censusTract` are assigned to each claim: *“U.S. Census Bureau defined census tracts; statistical subdivisions of a county or equivalent entity that are updated prior to each decennial census. The NFIP relies on our geocoding service to assign 11-digit census tract codes.”* Since FEMA states outright that its own geocoding service, not the claimant, assigns tract codes, and tract and block-group codes are two views of the same lookup, a single geocoding service assigning a recent-release boundary at processing time is the most direct reading of why block-group match rates follow loss year the way they do.

This is exactly the two-release cutover `default` implements for block group and why `default` uses a single cutover year (2020) rather than a historical loss-year rule.

Table 2 gives the full `block_group_match_status` breakdown across the CONUS-filtered pluvial population (1,155,473 claims, Table 1), evaluated using the `default` strategy before the final triangulation-overlap drop that produces the 1,149,288-claim released file. This population is intentionally broader than the released file, so that a source’s own match quality is not conflated with whether the other two sources happened to combine into a non-empty triangulated polygon.

<code>block_group_match_status</code>	<i>n</i> claims	%
<code>validated</code> (matched, polygon overlaps lat/lon box)	1,115,948	96.58%
<code>not_found</code> (code not in shapefile)	38,127	3.30%
<code>spatially_inconsistent</code> (matched code, polygon does not overlap lat/lon box)	1,398	0.12%

Table 2: `block_group_match_status` breakdown across the CONUS-filtered pluvial population (1,155,473 claims, Table 1), evaluated before the final triangulation-overlap drop — this is not the released file’s row count (1,149,288). See Data Records for status definitions.

ZCTA (Zip Code Tabulation Area) boundaries. ZCTAs exist officially only from the 2000 census onward. Unlike block group, whose `default` strategy uses a single 2010/2020 cutover, ZCTA has three configured releases (2000, 2010, 2020) and no fixed cutover year at all. Every claim’s loss year, including years well after 2000, is matched under `default` to whichever of the three is numerically closest, so a claim with loss year 2015 is tested against the 2010 release and a claim with loss year 2022 against the 2020 release, continuously rather than at a single switchover point. Most of this population’s claims carry a well-formed five-digit `reportedZipCode` and are eligible to be tested against ZCTA boundaries.

Within this eligible set, match rates are high and far more stable across releases than for block group, ranging 95.6–99.5% for the 2000 release and 98.1–100% for the 2010 and 2020 releases across every loss year (Table 5). `default`, equivalent to `closest` over the configured releases {2000, 2010, 2020}, matches 99.20% of these claims. Testing all three available ZCTA releases per claim and keeping the best match recovers 4,749 additional matches (0.41%) over `default` alone, a small gap compared to the double-digit percentage-point differences between releases seen in the block-group case above.

This stability is consistent with how `reportedZipCode` is sourced. Unlike `censusBlockGroupFips` and `censusTract`, `reportedZipCode` is not assigned by FEMA’s geocoding service. FEMA’s own data dictionary describes it as raw data “*as reported by WYO partners*,” recorded once at claim intake rather than reassigned by a lookup that could shift with whichever boundary release happens to be current at processing time. With no processing-time geocoding step to introduce a release-dependent pattern, there is no equivalent mechanism to produce the block-group-style crossover, consistent with the near-flat match rates observed across all three ZCTA releases. Furthermore, the `closest` strategy aligns with existing work that joins NFIP claims to external sources (e.g. the American Community Survey) [2, 3] and thus we elect this strategy as the default for ZCTA.

However, as with `block_group_strategy`, the `zcta_strategy` parameter remains configurable.

Table 3 gives the full `zip_match_status` breakdown across the same CONUS-filtered pluvial population (1,155,473 claims), evaluated under the `default` strategy before the same final triangulation-overlap drop described above.

<code>zip_match_status</code>	<i>n</i> claims	%
<code>validated</code> (matched, polygon overlaps lat/lon box)	1,130,162	97.81%
<code>not_found</code> (code not in shapefile)	16,226	1.40%
<code>spatially_inconsistent</code> (matched code, polygon does not overlap lat/lon box)	9,085	0.79%

Table 3: `zip_match_status` breakdown across the CONUS-filtered pluvial population (1,155,473 claims, Table 1), evaluated before the final triangulation-overlap drop — this is not the released file’s row count (1,149,288). See Data Records for status definitions.

Loss year	N claims	1990		2000		2010		2020	
		Match	Valid	Match	Valid	Match	Valid	Match	Valid
1978	12,571	71.8	97.7	36.5	99.2	100.0	100.0	80.9	99.1
1979	34,607	38.9	95.7	20.5	98.5	100.0	100.0	77.9	99.1
1980	28,306	71.3	97.9	34.8	98.7	100.0	100.0	80.2	99.2
1981	17,225	42.7	97.3	19.9	99.1	100.0	100.0	76.9	98.8
1982	21,415	63.8	96.2	30.6	98.4	100.0	100.0	80.8	98.8
1983	25,558	60.9	96.7	30.5	99.0	100.0	100.0	74.2	98.0
1984	6,671	65.4	96.2	25.0	98.5	100.0	100.0	82.3	99.3
1985	1,099	55.0	97.2	21.9	98.8	100.0	99.8	68.2	98.1
1986	1,846	51.7	95.4	25.4	98.9	100.0	100.0	75.1	98.1
1987	3,769	53.1	96.7	27.8	99.2	100.0	100.0	77.7	99.0
1988	6,690	64.0	97.7	41.9	98.1	100.0	100.0	78.6	98.9
1989	18,104	43.5	96.7	23.2	97.9	99.8	100.0	70.8	98.1
1990	7,074	67.6	96.3	32.3	99.0	99.9	100.0	79.8	98.7
1991	15,535	61.5	94.8	27.2	96.2	99.8	100.0	74.0	98.2
1992	22,375	44.4	95.3	25.6	96.5	99.7	100.0	78.8	98.9
1993	9,906	52.9	91.4	24.7	98.7	99.8	100.0	76.1	97.2
1994	15,283	41.6	95.4	17.9	98.0	99.8	100.0	73.4	97.8
1995	52,163	65.5	97.8	39.8	99.4	99.2	100.0	67.7	99.0
1996	21,212	53.8	95.8	18.4	97.5	99.7	100.0	79.3	98.5
1997	16,636	53.3	93.3	21.9	97.9	99.5	100.0	73.4	98.0
1998	44,433	48.0	93.5	19.5	97.7	99.5	100.0	69.8	98.1
1999	38,577	29.2	96.4	22.2	98.3	99.3	100.0	73.0	98.6
2000	14,785	22.3	95.2	30.3	98.1	99.0	100.0	61.5	98.8
2001	41,156	23.4	95.4	16.9	97.2	99.2	100.0	68.6	98.1
2002	22,192	54.6	94.3	28.3	98.1	99.4	100.0	68.5	98.0
2003	22,358	49.3	95.5	23.2	96.3	99.0	99.9	76.8	97.4
2004	29,718	42.3	94.4	19.5	97.4	98.3	99.9	72.7	97.8
2005	118,823	62.6	96.6	40.1	98.8	97.2	100.0	69.6	98.5
2006	18,477	51.1	96.0	20.6	96.7	98.4	99.9	73.8	98.4
2007	15,925	61.9	96.0	19.1	97.3	99.0	99.9	80.5	98.6
2008	46,008	33.5	92.4	12.7	97.5	98.7	99.9	71.2	96.3
2009	19,484	38.4	95.3	19.2	97.3	97.9	100.0	70.7	98.0
2010	18,842	61.6	96.1	27.3	97.3	98.6	99.9	81.4	98.4
2011	38,959	63.6	96.4	22.3	98.1	97.8	99.9	86.4	99.0
2012	52,844	54.8	94.2	25.7	95.8	94.4	99.8	84.0	99.4
2013	9,346	53.2	96.1	29.0	98.5	95.0	99.8	77.4	98.5
2014	9,688	47.4	96.2	22.8	98.7	95.3	99.8	77.8	98.9
2015	16,323	34.1	95.5	16.9	98.1	93.3	99.9	74.8	98.5
2016	30,840	40.3	95.9	25.0	97.1	84.8	99.6	71.8	98.3
2017	51,998	21.4	91.9	8.0	95.8	84.3	99.5	76.5	98.4
2018	11,243	39.9	95.8	14.9	98.0	87.5	99.5	82.7	98.7
2019	15,159	40.2	89.5	13.1	96.4	87.2	99.5	79.5	98.5
2020	15,237	35.2	93.9	19.4	98.1	80.2	99.7	83.2	99.2
2021	26,083	54.9	97.0	20.7	97.7	82.4	99.4	88.7	99.5
2022	21,247	30.3	85.0	24.6	86.1	76.9	95.5	92.5	98.9
2023	12,441	39.9	95.8	18.5	97.6	79.8	99.5	94.9	99.7
2024	47,167	26.1	93.2	23.8	95.7	75.1	98.5	98.8	99.9
2025	7,424	36.8	95.2	13.0	95.0	58.1	98.2	99.7	100.0
2026	651	41.9	94.1	19.1	96.8	72.0	98.3	100.0	100.0

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(continued)

Loss year	N claims	1990		2000		2010		2020	
		Match	Valid	Match	Valid	Match	Valid	Match	Valid

Table 4: Block-group GEOID match rate and spatial-validation rate (%) by loss year and shapefile release, for pluvial (`causeOfDamage == "4"`) claims. Match is the raw GEOID match rate against that release's shapefile. Valid is the share of matched claims whose polygon also overlaps the claim's rounded lat/lon box. For each loss year, the highest Match value and the highest Valid value across releases are shown in bold (ties bolded in full). Shaded cells mark the release the `default` strategy selects for that loss year.

Loss year	N claims	2000		2010		2020	
		Match	Valid	Match	Valid	Match	Valid
1978	12,266	99.4	99.2	99.7	99.1	99.7	99.1
1979	33,519	98.9	98.8	99.3	98.7	99.3	98.7
1980	27,439	99.2	99.2	99.6	99.2	99.6	99.2
1981	16,583	99.0	98.7	99.6	98.5	99.7	98.5
1982	20,354	97.6	99.0	98.9	98.7	98.8	98.7
1983	23,718	98.9	98.5	99.4	98.4	99.4	98.5
1984	6,198	98.9	98.8	99.2	98.9	99.2	99.0
1985	1,071	97.8	98.5	98.2	98.3	98.2	98.2
1986	1,824	98.1	98.4	99.1	98.4	99.1	98.5
1987	3,745	98.2	98.6	99.0	98.7	99.0	98.7
1988	6,670	97.7	99.0	98.1	99.0	98.1	99.0
1989	18,056	99.0	98.1	99.3	98.1	99.3	98.0
1990	7,036	98.7	98.2	99.1	98.2	99.1	98.2
1991	15,491	98.2	98.4	98.8	98.4	98.8	98.4
1992	22,311	98.2	98.5	99.0	98.5	99.0	98.5
1993	9,851	97.8	97.6	98.3	97.4	98.4	97.4
1994	15,253	98.8	98.2	99.2	98.2	99.2	98.2
1995	52,106	99.0	98.8	99.5	98.7	99.6	98.7
1996	21,156	97.6	98.5	98.7	98.4	98.8	98.4
1997	16,610	98.6	98.7	99.4	98.7	99.5	98.7
1998	44,393	98.0	98.2	99.3	98.1	99.3	98.1
1999	38,556	98.9	99.4	99.7	99.3	99.7	99.3
2000	14,783	99.5	99.3	99.7	99.3	99.8	99.3
2001	41,137	98.9	99.1	99.6	99.1	99.6	99.1
2002	22,186	99.1	99.1	99.5	99.1	99.5	99.1
2003	22,353	98.6	99.1	99.5	99.2	99.6	99.1
2004	29,707	96.0	99.3	99.4	99.3	99.5	99.3
2005	118,800	99.3	99.6	99.8	99.6	99.8	99.6
2006	18,470	98.2	99.5	99.5	99.5	99.5	99.4
2007	15,918	98.3	99.6	99.7	99.7	99.7	99.6
2008	45,993	99.1	97.3	99.7	97.3	99.8	97.3
2009	19,482	98.8	99.7	99.8	99.8	99.8	99.8
2010	18,841	98.8	99.7	99.8	99.7	99.9	99.7
2011	38,955	98.8	99.7	99.7	99.7	99.8	99.7
2012	52,837	98.8	99.7	99.6	99.8	99.6	99.8
2013	9,345	98.2	99.8	99.7	99.8	99.8	99.8
2014	9,688	95.6	99.8	99.9	99.8	100.0	99.8
2015	16,323	98.4	99.8	99.8	99.8	99.9	99.8
2016	30,839	98.2	99.9	99.9	99.9	99.9	99.9

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(continued)

Loss year	N claims	2000		2010		2020	
		Match	Valid	Match	Valid	Match	Valid
2017	51,998	97.7	99.8	100.0	99.8	100.0	99.8
2018	11,243	96.8	99.9	99.2	99.9	99.9	99.9
2019	15,158	98.9	99.8	99.6	99.9	100.0	99.9
2020	15,237	95.9	99.6	99.2	99.6	100.0	99.6
2021	26,063	98.8	99.8	99.8	99.9	99.9	99.9
2022	21,247	97.3	99.8	100.0	99.9	100.0	99.9
2023	12,439	98.7	99.8	99.8	99.8	99.9	99.8
2024	47,152	98.4	99.8	99.9	99.9	100.0	99.8
2025	7,421	97.4	99.7	99.4	99.8	99.9	99.8
2026	651	97.7	100.0	99.7	100.0	100.0	100.0

Table 5: ZCTA GEOID match rate and spatial-validation rate (%) by loss year and shapefile release, for pluvial (`causeOfDamage == "4"`) claims. Match is the raw GEOID match rate against that release’s shapefile. Valid is the share of matched claims whose polygon also overlaps the claim’s rounded lat/lon box. For each loss year, the highest Match value and the highest Valid value across releases are shown in bold (ties bolded in full). Shaded cells mark the release the `default` strategy selects for that loss year.

Extent of location refinement. Table 6 shows how many of the three location sources (block group, ZIP, lat/lon box) contributed to each claim’s triangulated polygon across the released file’s 1,149,288 claims. The large majority of claims combine all three sources into the final intersection.

n_geometry_sources	n claims	%
3	1,084,753	94.38%
2	64,234	5.59%
1	301	0.03%

Table 6: Number of contributing location sources per claim’s triangulated polygon, across the released file’s 1,149,288 claims.

geometry_sources	n claims	%
block_group+zip+latlon	1,084,753	94.38%
zip+latlon	39,224	3.41%
block_group+latlon	25,010	2.18%
latlon	301	0.03%

Table 7: Number of claims by the specific combination of contributing location sources, across the released file’s 1,149,288 claims. The lat/lon box contributes to every claim, so no combination omits it.

Figure 1 (top) and Table 8 show the resulting distribution of triangulated `polygon_area` across the released file’s full 1,149,288 claims, all of which fall within the 0–115 km² range shown. 910,194 claims (79.2%) resolve to a polygon of 5 km² or smaller, and the distribution is heavily right-skewed, with claim counts falling off sharply past 20 km². The median triangulated polygon area is 1.17 km², and the mean is 5.21 km², pulled well above the median by this right tail.

Figure 1 (bottom) and Table 9 show, for each claim, how the final triangulated `polygon_area` compares to the area of the 0.1°×0.1° lat/lon box source, computed as $1 - (\text{polygon_area} / \text{lat-lon box area})$ and expressed

as a percentage. Across the released file’s 1,149,288 claims, 923,947 (80.4%) show a reduction of at least 94.7%, and the distribution is heavily left-skewed toward this upper end. The mean reduction is 94.9%, and the median is 98.9%.

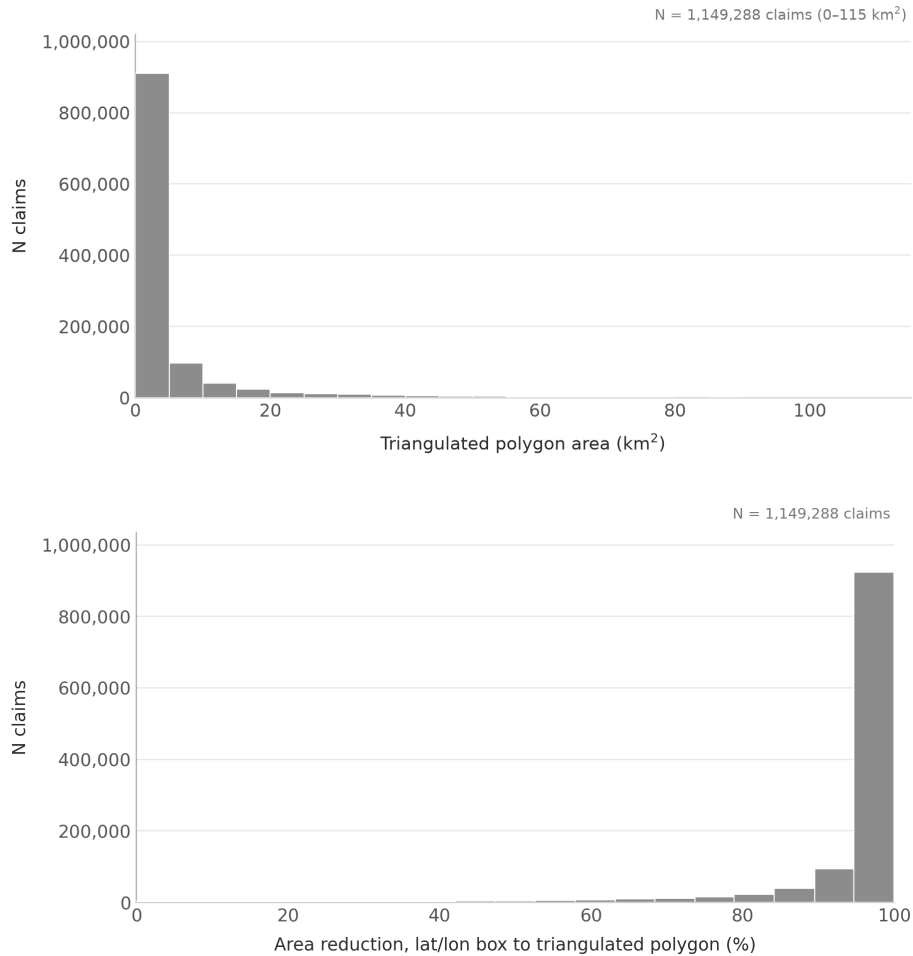


Figure 1: Top, distribution of triangulated `polygon_area` (km²) across the released file’s claims. Bottom, distribution of the fractional area reduction between the 0.1°×0.1° lat/lon box source area and the final triangulated `polygon_area`, across the released file’s claims.

Pluvial date correction

Date-correction outcomes. Table 10 gives the full `pluvialCorrectionStatus` breakdown across the released file’s 1,149,288 claims, under the default 7.62 mm constant precipitation threshold. `no_aorc_data_in_window` arises, for example, when a claim sits on a small land mass whose four surrounding AORC grid points for bilinear interpolation are all ocean pixels with no recorded precipitation.

Pluvial precipitation threshold. Figure 2 and Table 11 show the distribution of `reportedDateMaxPrecipMm`, the maximum hourly precipitation at each pluvial claim’s triangulated location on the reported `dateOfLoss`, before any correction is applied. This value is populated only when AORC has valid data on that specific date, a narrower condition than the one `pluvialCorrectionStatus` is decided under, which instead uses the best precipitation found anywhere across the full ± 7 -day search window. A claim can therefore still be `corrected` or `no_qualifying_precip_in_window` even when AORC is missing data on the reported date itself, so long as some other day in the window has data, leaving `reportedDateMaxPrecipMm` undefined for

Bin (km ²)	<i>n</i> claims	%	Bin (km ²)	<i>n</i> claims	%
0–5	910,194	79.2%	60–65	2,829	0.2%
5–10	97,544	8.5%	65–70	2,349	0.2%
10–15	40,798	3.5%	70–75	2,232	0.2%
15–20	24,614	2.1%	75–80	1,585	0.1%
20–25	14,917	1.3%	80–85	900	0.1%
25–30	11,324	1.0%	85–90	1,267	0.1%
30–35	9,667	0.8%	90–95	714	0.1%
35–40	7,565	0.7%	95–100	1,069	0.1%
40–45	6,218	0.5%	100–105	694	0.1%
45–50	4,358	0.4%	105–110	766	0.1%
50–55	4,231	0.4%	110–115	58	0.0%
55–60	3,395	0.3%			

Table 8: Triangulated `polygon_area` bin counts (km²) across the released file’s 1,149,288 claims. All fall within the 0–115 km² range shown.

Bin (%)	<i>n</i> claims	%	Bin (%)	<i>n</i> claims	%
0.0–5.3	1,959	0.2%	52.6–57.9	5,820	0.5%
5.3–10.5	1,268	0.1%	57.9–63.2	6,792	0.6%
10.5–15.8	894	0.1%	63.2–68.4	9,839	0.9%
15.8–21.1	1,382	0.1%	68.4–73.7	11,287	1.0%
21.1–26.3	1,265	0.1%	73.7–78.9	14,909	1.3%
26.3–31.6	2,718	0.2%	78.9–84.2	21,492	1.9%
31.6–36.8	2,381	0.2%	84.2–89.5	39,238	3.4%
36.8–42.1	2,826	0.2%	89.5–94.7	92,900	8.1%
42.1–47.4	4,048	0.4%	94.7–100.0	923,947	80.4%
47.4–52.6	4,323	0.4%			

Table 9: Bin counts of the fractional area reduction between the 0.1°×0.1° lat/lon box source area and the final triangulated `polygon_area`, across the released file’s 1,149,288 claims.

24,830 of the released file’s 1,149,288 claims. Of these, 12,425 are `before_aorc_coverage` (never evaluated) and 12,204 are `no_aorc_data_in_window` (no day in the window has data, the reported date included), while the remaining 201 are `corrected` or `no_qualifying_precip_in_window` claims resolved from a different day in the window than the one missing data. Of the 1,124,458 claims with a defined `reportedDateMaxPrecipMm`, 362,174 (32.21%) fall at or below the 7.62 mm default threshold, the population eligible to have their date corrected under the default constant threshold.

Under the default uniform 7.62 mm threshold, 210,488 of 1,149,288 claims (18.3%) have a `correctedDateOfLoss` that differs from the originally reported `dateOfLoss`. Figure 3 and Table 12 show how the full `pluvialCorrectionStatus` breakdown responds to the constant precipitation threshold, swept over six values (2.5, 5, 7.62, 10, 15, and 20 mm) with the search window otherwise unchanged. The `corrected` share is not monotonic in the threshold. It rises from 14.3% at 2.5 mm to a peak of 19.4% at 10 mm, then falls to 17.8% at 20 mm, reflecting two opposing effects as the threshold rises. The `accepted_as_reported` share falls monotonically (81.0% to 36.6%, since fewer reported dates clear a higher bar on their own), while the `no_qualifying_precip_in_window` share rises monotonically (2.5% to 43.4%, since fewer search windows contain any day clearing a higher bar). The `corrected` share is what remains between these two trends at each threshold. `no_aorc_data_in_window` and `before_aorc_coverage` are each unaffected by the threshold and together account for a constant 2.1% of claims across the sweep.

Figure 4 and Table 13 show, for the same six thresholds, the distribution of the resulting shift in `correctedDateOfLoss` away from the originally reported `dateOfLoss`. The figure covers `corrected` claims only, with each thresh-

old’s bars normalized by that threshold’s own **corrected** count. A shift of 0 days does not occur in this population by construction, and the ± 7 -day search window bounds every shift to $[-7, 7]$. At every threshold tested, the shift is concentrated in a small number of days. The share of corrected claims shifting by no more than 2 days ranges from 49.1% at 2.5 mm to 68.4% at 20 mm, and a shift of exactly -1 day (a qualifying day immediately before the reported date) is the single most common outcome at every threshold except 20 mm, where $+1$ day is marginally more common. The table instead shares one denominator, the released file’s full claim count, across every column and every threshold, so that the columns can be read as a decomposition of the same population. Its 0 column holds **accepted_as_reported** claims, for which **correctedDateOfLoss** equals **dateOfLoss** by definition, alongside the ± 1 through ± 7 columns of **corrected** claims. **no_qualifying_precip_in_window** and no-AORC-data claims are never assigned a shift and so fall outside the table.

pluvialCorrectionStatus	<i>n</i> claims	%	Why
accepted_as_reported	762,284	66.33%	Reported date already coincided with meaningful precipitation
corrected	210,488	18.31%	A qualifying day was found within the search window
no_qualifying_precip_in_window	151,887	13.22%	AORC data was available, but no day in the window cleared the threshold
before_aorc_coverage	12,425	1.08%	dateOfLoss predates AORC’s 1979-02-01 start, never evaluated
no_aorc_data_in_window	12,204	1.06%	Evaluated, but every hour of every day in the search window was missing from AORC

Table 10: **pluvialCorrectionStatus** breakdown across the full released population, under the default 7.62 mm constant precipitation threshold. For **accepted_as_reported** and **corrected**, **correctedDateOfLoss** reflects a day of qualifying precipitation. For the remaining statuses it equals the originally reported date, per its always-populated fallback behavior described in Data Records.

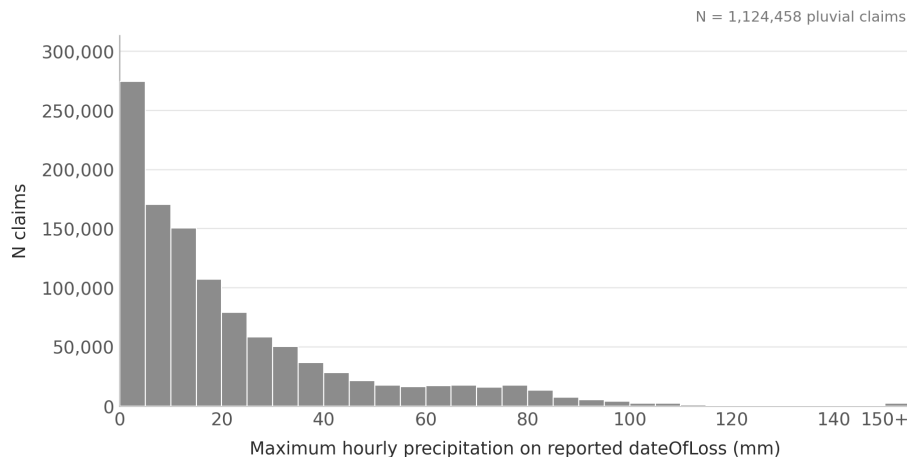


Figure 2: Distribution of **reportedDateMaxPrecipMm** across the 1,124,458 pluvial claims with a defined **reportedDateMaxPrecipMm**.

Bin (mm)	<i>n</i> claims	%	Bin (mm)	<i>n</i> claims	%
0–5	274,843	24.4%	80–85	13,515	1.2%
5–10	170,513	15.2%	85–90	7,533	0.7%
10–15	150,730	13.4%	90–95	5,483	0.5%
15–20	107,410	9.6%	95–100	4,253	0.4%
20–25	79,226	7.0%	100–105	2,588	0.2%
25–30	58,470	5.2%	105–110	2,301	0.2%
30–35	50,487	4.5%	110–115	1,261	0.1%
35–40	36,867	3.3%	115–120	596	0.1%
40–45	28,181	2.5%	120–125	511	0.0%
45–50	21,384	1.9%	125–130	650	0.1%
50–55	17,603	1.6%	130–135	625	0.1%
55–60	16,471	1.5%	135–140	791	0.1%
60–65	17,392	1.5%	140–145	615	0.1%
65–70	17,623	1.6%	145–150	760	0.1%
70–75	15,914	1.4%	150+	2,303	0.2%
75–80	17,559	1.6%			

Table 11: `reportedDateMaxPrecipMm` bin counts across the 1,124,458 pluvial claims with a defined `reportedDateMaxPrecipMm`.

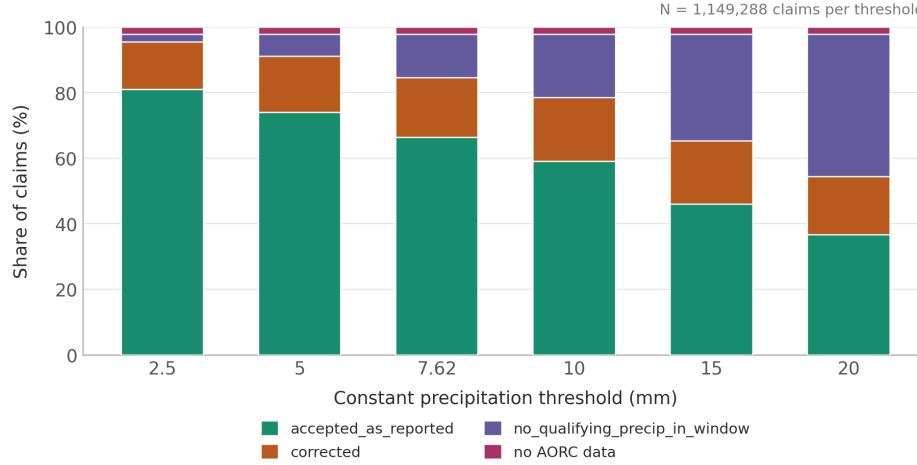


Figure 3: `pluvialCorrectionStatus` breakdown across the released file’s claims, swept over six constant precipitation thresholds (2.5, 5, 7.62, 10, 15, and 20 mm; 7.62 mm is the released default). `no_aorc_data` combines `no_aorc_data_in_window` with the threshold-invariant `before_aorc_coverage`.

Threshold (mm)	Accepted as Reported	Corrected	No Qualifying Precip in Window	No AORC Data
2.5	81.0%	14.3%	2.5%	2.1%
5	73.9%	17.2%	6.7%	2.1%
7.62*	66.3%	18.3%	13.2%	2.1%
10	59.1%	19.4%	19.4%	2.1%
15	46.0%	19.3%	32.5%	2.1%
20	36.6%	17.8%	43.4%	2.1%

Table 12: `pluvialCorrectionStatus` share of the released file’s 1,149,288 claims, swept over six constant precipitation thresholds. Columns correspond to `accepted_as_reported`, `corrected`, `no_qualifying_precip_in_window`, and, combined, `no_aorc_data_in_window` with the threshold-invariant `before_aorc_coverage`. *7.62 mm (shaded) is the released default. Table 10 gives the full n claims breakdown at this default threshold.

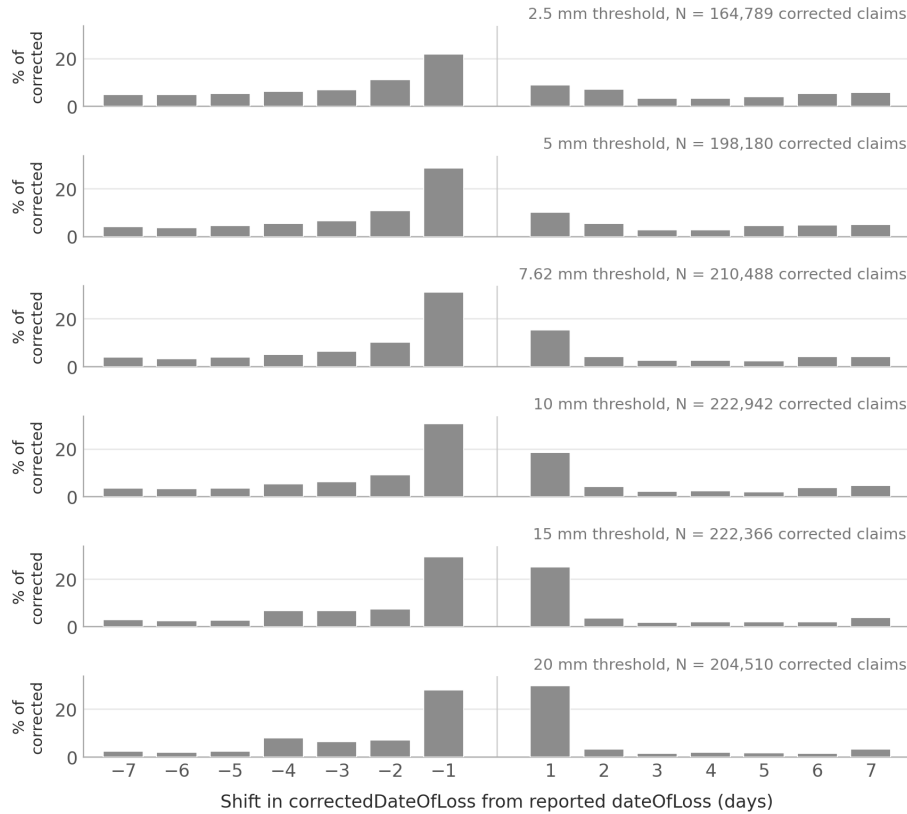


Figure 4: Distribution of the shift in `correctedDateOfLoss` away from the originally reported `dateOfLoss` (days, negative = earlier) among `corrected` claims, one row per constant precipitation threshold swept (2.5, 5, 7.62, 10, 15, and 20 mm; 7.62 mm is the released default).

Scope of validation

This dataset’s claim population follows FEMA’s own `causeOfDamage` coding, which this pipeline takes as given rather than re-deriving. Prior work has found NFIP’s reported cause-of-loss information to be incomplete or inconsistent in places [11]; because this pipeline builds directly on the `causeOfDamage` field, it inherits that limitation. The spatial and temporal refinements validated above apply to whichever claims

Threshold (mm)	-7	-6	-5	-4	-3	-2	-1	0	+1	+2	+3	+4	+5	+6	+7
2.5	0.7%	0.7%	0.8%	0.9%	1.0%	1.6%	3.1%	81.0%	1.3%	1.0%	0.5%	0.5%	0.6%	0.8%	0.8%
5	0.7%	0.7%	0.8%	1.0%	1.1%	1.9%	4.9%	73.9%	1.8%	1.0%	0.5%	0.5%	0.8%	0.8%	0.9%
7.62*	0.7%	0.6%	0.7%	0.9%	1.2%	1.9%	5.7%	66.3%	2.8%	0.8%	0.5%	0.5%	0.5%	0.8%	0.8%
10	0.7%	0.6%	0.7%	1.0%	1.2%	1.8%	5.9%	59.1%	3.6%	0.8%	0.5%	0.5%	0.4%	0.7%	0.9%
15	0.6%	0.5%	0.6%	1.3%	1.3%	1.5%	5.7%	46.0%	4.9%	0.7%	0.4%	0.4%	0.4%	0.4%	0.8%
20	0.4%	0.4%	0.4%	1.4%	1.2%	1.3%	5.0%	36.6%	5.3%	0.6%	0.3%	0.3%	0.3%	0.3%	0.6%

Table 13: Share of the released file’s 1,149,288 claims by shift in `correctedDateOfLoss` away from the originally reported `dateOfLoss` (days, negative = earlier), swept over six constant precipitation thresholds, with every column sharing the same 1,149,288-claim denominator. *7.62 mm (shaded) is the released default. The 0 column holds `accepted_as_reported` claims, for which `correctedDateOfLoss` equals `dateOfLoss` by definition (shift = 0); the ± 1 through ± 7 columns hold `corrected` claims. Claims with `no_qualifying_precip_in_window` or no AORC data are never assigned a shift and so are outside this table’s scope, which is why rows do not sum to 100%. Table 12 gives the full `pluvialCorrectionStatus` breakdown per threshold.

FEMA has coded as pluvial at the time of the source extract.

Similarly, the boundary-release validation above takes each claim’s reported latitude/longitude as accurate and checks the other two location sources against it. It does not, and cannot from this data alone, establish that the reported latitude/longitude itself is accurate. Any location error already present in FEMA’s reported coordinates is a limitation this pipeline.

The pluvial date correction has the same character. No independent record of a claim’s true flood-triggering date exists against which `correctedDateOfLoss` could be checked. The threshold sweep and day-shift analysis above (Figures 3 and 4) characterize how the correction behaves as its parameters change, not whether the resulting dates are closer to the truth than the originally reported ones. The correction is therefore best understood as making each claim’s reported date more plausible under an explicit, stated prior (higher local rainfall intensity is taken as more consistent with a pluvial flood), not as a verified and absolute error correction.

Usage Notes

The full pipeline is run via `src/run_pipeline.sh` in the code repository, which executes the following six ordered scripts. Each script reads its inputs from the previous step’s output path, so once those outputs exist any single step can be re-run on its own without repeating the full pipeline.

1. `download_claims.py` — fetches the raw FEMA claims extract and the FRED inflation series.
2. `adjust_inflation.py` — Stage 1, inflation adjustment.
3. `triangulate_claims.py` — Stage 2, spatial triangulation.
4. `build_aorc_pixel_day_index.py` — indexes the AORC pixel/days needed for pluvial claims.
5. `fetch_aorc_daily_max.py` — fetches the corresponding AORC precipitation.
6. `correct_pluvial_dates.py` — Stage 3, pluvial date correction.

`run_pipeline.sh` restricts the run to pluvial claims (`causeOfDamage == "4"`) by default, matching this dataset’s release, but this is a run-time flag rather than a hardcoded choice. The methodological parameters described in Methods are all exposed as command-line flags.

- `triangulate_claims.py --block-group-strategy / --zcta-strategy` — the block-group/ZCTA boundary-release strategy (default, `closest`, `most_recent`, or `drop`).
- `triangulate_claims.py --cause-of-damage` — the `causeOfDamage` filter (`run_pipeline.sh` passes 4 for pluvial).

- `correct_pluvial_dates.py --min-hourly-precip-mm` — the constant precipitation threshold.
- `correct_pluvial_dates.py --threshold-netcdf` — a spatially varying threshold from a NetCDF file, in place of a constant.

Census block-group and ZCTA shapefiles must be downloaded separately and pointed to via `config.yaml`, while the FEMA, FRED, and AORC inputs are fetched automatically.

Data Availability

The derived dataset described in this paper is released under a CC-BY 4.0 license and is deposited on Zenodo at <https://doi.org/10.5281/zenodo.22019526>. The three upstream sources it is built from are also public, namely FEMA’s OpenFEMA FIMA NFIP Redacted Claims v2 (<https://www.fema.gov/openfema-data-page/fima-nfip-redacted-claims-v2>), U.S. Census Bureau cartographic boundary files (<https://www.census.gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html>), and NOAA’s AORC v1.1 (public S3, `s3://noaa-aws-aorc-v1-1-1km`, no-sign-request). None impose redistribution restrictions beyond those already stated.

Code Availability

The pipeline used to construct the derived dataset is publicly available on GitHub at <https://github.com/yuya737/nfip-pluvial-claims-refinement>, released under the MIT license.

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